Experiment 1(a): 8-Bit Addition with Carry Using Direct Addressing

AIM:

To write an assembly language program to add two 8-bit numbers stored in memory locations 4200H and 4201H and store the result in 4202H and carry in 4203H using direct addressing.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

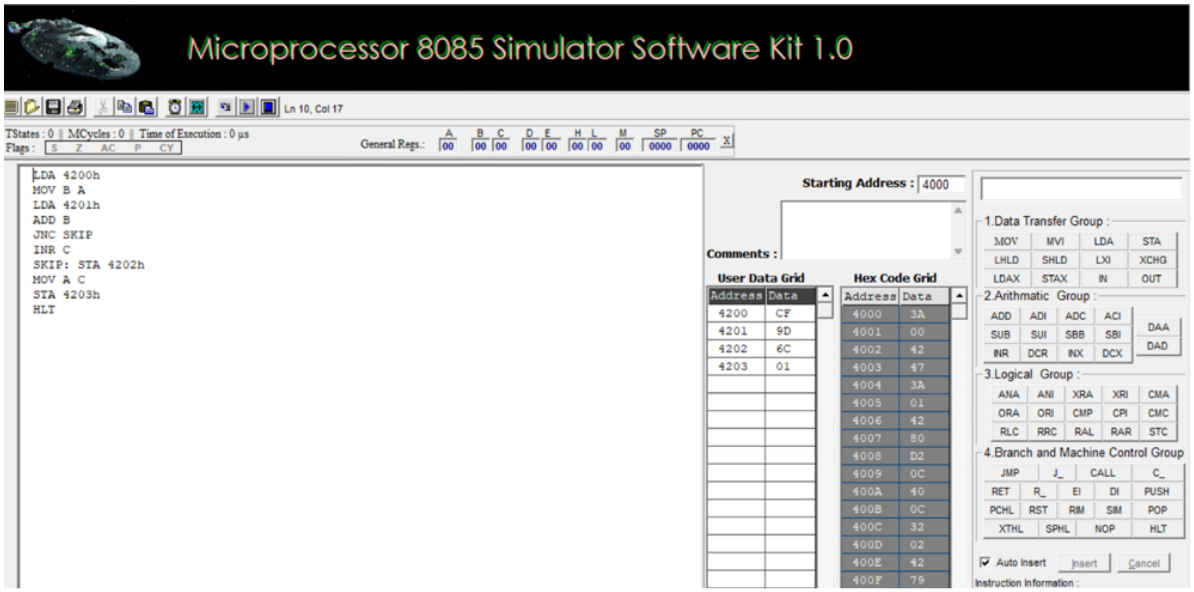
ALGORITHM:

1. Load the first number from memory into the accumulator.
2. Move the accumulator content to register B.
3. Load the second number into the accumulator.
4. Add the content of register B to the accumulator.
5. Store the sum in memory location 4202H.
6. Store the carry in memory location 4203H.
7. Stop the program.

PROGRAM:

```
LDA 4200H
MOV B,A
LDA 4201H
ADD B
STA 4202H
MVI A,00H
ADC A
STA 4203H
HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program for adding two 8-bit numbers with carry using direct addressing was written and executed successfully using the 8085 microprocessors.